

- Make the necessary property changes like we did in changing the text property, for the label and button components.
- Now your graphical user interface is ready. Figure 4 shows exactly what the application will look like after the installation.

Now let's head towards the block editor to define the features, so that we can determine the actual functionality that we expect from our application.

- When we press the 'Start' button, the counter value should increase and continue to do so.
 - When we press the 'Stop' button, the counter value should be paused to the current value.
 - When we press the 'Reset' button, the counter value should be reset to the default 0.
- So let's move on and add these two features using the block editor.

Block editor blocks

- The blocks shown in Figure 7 are easy to understand. These say that when the button 'Start' is pressed, it should increase whatever figure is visible as the count label.
- When the button 'Stop' is pressed, it should stop the timer at the current value. For stopping the clock, we do not need to set it to 'disable'; instead, we need to set 'enable' as false.
- When the 'Reset' button is pressed, it stops the clock and sets the counter value to the default 0.
- The fourth block is the one that you will see for the very first time. It indicates that when the timer is enabled, it will increase the label value by one in each repeat cycle. If you remember the properties of the clock component, we have set the timer interval as 1000 milliseconds, which means the clock will have a new cycle every second and hence will be increasing the value.

Now you are done with the block editor too. Next, we will move on to download and install the app on your phone to check how it works.

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

- On the top row, click on the 'Build' button. It will show you the option to download the apk to your computer.
- As the downloading is going on, the progress can be seen and after a successful download, the application will be placed in the download folder of your directory or in the location you have set for it.
- Now you need to get this apk file to your mobile phone either via Bluetooth or USB cable. Once you have placed the apk file on your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying, "Installing from un-trusted



Figure 4: Automatic counter designer screen



Figure 5: Automatic counter components

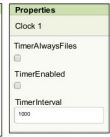


Figure 6: Properties for clock component



Figure 7: Automatic counter block editor blocks

source.” Allow this from the *Settings* and after successful installation, you will see the icon of your application in the menu of your mobile. Here, you will see the default icon, which can be changed (we will tell you how to do so as we move ahead).

I hope your application is working exactly as per your requirements. Now, depending upon your usability and customisation, you can change various things—buttons, text and behaviour too.

Debugging the application

As of now, we have just created the prototype of the application with very basic functionality. What else would a user be interested in? Take care that certain features of the app do not annoy the user and check if your app is able to sustain itself. Consider the following cases:

- Don't you think there should be a reset button to clear the value and start counting from the beginning?
- If you were asked to add an additional reset button, will you be able to do so?
- What extra GUI feature do you think should be added?
- How many blocks should be added to furnish the reset functionality?
- Will it be useful to store the counter value in the database and retrieve it if needed?

These are some of the scenarios that might occur and users will be pretty happy seeing them implemented.

Think over all the needs that may crop up, and how you could integrate these into the application. Do revert me if you fail to accomplish any of the above features. 

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